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Vehicle battery protection structure

Abstract

A vehicle-body structure improves mounting rigidity of a battery protection frame and rigidity of the battery protection frame itself by including a battery protection frame which has a hollow cross-section, a vehicle-body-side frame which extends along the battery protection frame and is coupled with the battery protection frame, and a fixation member which is inserted through a hole portion of the battery protection frame and is capable of fixing the battery protection frame to a vehicle body component as another component than the vehicle-body-side frame in an internal portion of the battery protection frame.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) The present application claims priority to Japanese application number 2022-026784 filed in the Japanese Patent Office on Feb. 24, 2022, the entire contents of which are incorporated herein by reference.

FIELD

(2) The present disclosure relates to a vehicle-body structure for an electric vehicle, for example.

DESCRIPTION OF THE RELATED ART

(3) For example, Patent Literature 1 discloses a vehicle in which batteries are installed below a floor panel. In Patent Literature 1, the vehicle includes an annular vehicle-body bottom skeletal frame which is formed along an outer periphery of the floor panel and an annular battery installation frame which is fixed to the vehicle-body bottom skeletal frame, and the batteries are arranged on the inside of the battery installation frame. The battery installation frame is provided with a mounting upper wall and a mounting lower wall so as to form a flange which is protruded outward, and those mounting upper wall and mounting lower wall are arranged on a lower surface of the vehicle-body bottom skeletal frame and are fastened thereto by bolts.

CITATION LIST

Patent Literature

(4) [Patent Literature 1] International Publication No. WO 2020/105283

SUMMARY

Problems to be Solved

(5) Incidentally, in a case of an electric vehicle, because an installed amount of batteries for supplying electric power to a traveling motor directly influences a cruisable distance, it is demanded that the installed amount of the batteries be increased as much as possible. In this regard, in Patent Literature 1, because a mounting upper wall and a mounting lower wall which are protruded outward from a battery installation frame are arranged on and fastened to a lower surface of a vehicle-body bottom skeletal frame, an installation space for the batteries has to be made narrower by spaces for protruded parts of the mounting upper wall and the mounting lower wall.

(6) Accordingly, it is possible that the mounting upper wall and the mounting lower wall in Patent Literature 1 are omitted and the battery installation frame is thereby directly fastened to a vehicle body. Consequently, it becomes possible to increase the installed amount by enlarging the installation space for the batteries by the spaces for the omitted mounting upper wall and mounting lower wall.

(7) However, when the mounting upper wall and the mounting lower wall become absent, how to secure mounting rigidity of the battery installation frame on the vehicle body becomes a problem, and battery protection performance might become insufficient due to lowering of rigidity of the battery installation frame itself.

(8) The present disclosure has been made in consideration of such problems, and one of the objects thereof is to enhance mounting rigidity of a battery protection frame on a vehicle body and to enhance rigidity of the battery protection frame itself as well.

Solutions to the Problems

(9) A first aspect of the present disclosure can be based on a vehicle-body structure for an electric vehicle which includes a traveling motor and in which a battery supplying electric power to the traveling motor is disposed below a floor panel. The vehicle-body structure includes: a battery protection frame that protects the battery and has a hollow cross-section; a vehicle-body-side frame that extends toward the battery protection frame and is coupled with the battery protection frame; and a fixation member that is inserted into a space of an internal portion of the battery protection frame through a hole portion opening in an outer surface of the battery protection frame and connected with the space, and that is capable of fixing the battery protection frame to a vehicle body component as another component than the vehicle-body-side frame in the internal portion of the battery protection frame.

(10) With this configuration, because the battery protection frame has the hollow cross-section, the battery protection frame is provided which is highly rigid and light. Further, because the vehicle-body-side frame that extends along the battery protection frame is coupled with the battery protection frame, this also enhances rigidity of the battery protection frame.

(11) Furthermore, because the battery protection frame is, in its internal portion, fixed to the

vehicle body component by the fixation member separately from the vehicle-body-side frame, a mounting upper wall and a mounting lower wall which are protruded outward as in related art are not necessary, and by their spaces, an installation space for the battery becomes wider.

(12) Further, because the battery protection frame is fixed in its internal portion, compared to a case where the battery protection frame is fixed to the vehicle body component in an outer portion of the battery protection frame, the distance between a portion on which a fixing force of the fixation member is exerted in the battery protection frame and the vehicle body component becomes short. Accordingly, mounting rigidity of the battery protection frame on a vehicle body is enhanced.

(13) The vehicle-body-side frame according to a second aspect of the present disclosure may be provided with a cover portion that is formed to extend along the outer surface, in which the hole portion is formed, in the battery protection frame and that covers the hole portion from an outside of the battery protection frame.

(14) With this configuration, entrance of foreign objects into the internal portion of the battery protection frame through the hole portion can be inhibited. Further, when the hole portion is left open, this might lead to lowering of aerodynamic performance in traveling, but with this configuration, because the hole portion is covered, lowering of aerodynamic performance in traveling can be avoided.

(15) The battery protection frame according to a third aspect of the present disclosure may be disposed below the vehicle body component. In this case, in the internal portion of the battery protection frame, a partition wall portion that marks off the space in an up-down direction can be provided between an upper wall portion and a lower wall portion of the battery protection frame, and the fixation member can exert a fixing force on the partition wall portion.

(16) With this configuration, the partition wall portion is provided, and rigidity of the battery protection frame itself can thereby further be enhanced. In this case, the fixing force is exerted on the partition wall portion by the fixation member, the distance between the portion on which the fixing force of the fixation member is exerted in the battery protection frame and the vehicle body component can thereby be made sufficiently short, and further, mounting rigidity of the battery protection frame on the vehicle body can sufficiently be enhanced.

(17) The vehicle-body-side frame according to a fourth aspect of the present disclosure may include a suspension connecting portion with which at least a part of a suspension apparatus is capable of being coupled. With this configuration, because the vehicle-body-side frame includes the suspension connecting portion, rigidity of the vehicle-body-side frame is enhanced. Because the highly rigid vehicle-body-side frame is coupled with the battery protection frame, rigidity of the battery protection frame can further be enhanced.

(18) The battery protection frame according to a fifth aspect of the present disclosure may extend in a vehicle width direction. The vehicle-body-side frame includes a lower plate portion that extends in the vehicle width direction along a lower wall portion of the battery protection frame and is fixed to the lower wall portion and a vertical plate portion that extends in the vehicle width direction along one wall portion of the battery protection frame in a vehicle front-rear direction and is fixed to the one wall portion.

(19) With this configuration, because the lower plate portion and the vertical plate portion of the vehicle-body-side frame can respectively be fixed to the lower wall portion of the battery protection frame and the one wall portion in the vehicle front-rear direction, the vehicle-body-side frame can firmly be fixed to the battery protection frame.

(20) The hole portion according to a sixth aspect of the present disclosure is formed in the lower wall portion of the battery protection frame. In this case, it becomes possible that the hole portion is covered from below by using the lower plate portion of the vehicle-body-side frame.

Advantages

(21) As described above, a vehicle-body-side frame is coupled with a battery protection frame which has a hollow cross-section, and rigidity of the battery protection frame itself can thereby be

enhanced. In addition, the battery protection frame is, in its internal portion, fixed to a vehicle body component, and mounting rigidity of the battery protection frame on a vehicle body can be enhanced as well.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a side view of an electric vehicle according to an embodiment, in which a part of the electric vehicle is omitted.
- (2) FIG. 2 is a side view illustrating a state where the electric vehicle is divided into a lower structure and an upper structure.
- (3) FIG. 3 is a plan view of the lower structure.
- (4) FIG. 4 is a bottom view of a rear-side portion of a vehicle-body structure.
- (5) FIG. 5 is a perspective view of a rear-side portion of the lower structure as seen from above.
- (6) FIG. 6 is a side view of the rear-side portion of the lower structure.
- (7) FIG. 7 is a bottom view of a rear-side portion of the upper structure.
- (8) FIG. 8 is a cross-sectional view in which a portion of the vehicle-body structure on a rear side and a left side is sectioned in a vertical direction.
- (9) FIG. 9 is an enlarged cross-sectional view in which the portion of the vehicle-body structure on the rear side and the left side is sectioned in the vertical direction.
- (10) FIG. 10 is a perspective cross-sectional view in which the portion of the vehicle-body structure on the rear side and the left side is sectioned in the vertical direction.

DETAILED DESCRIPTION

- (11) The present disclosure will hereinafter be described in detail based on drawings. Note that the description in the following is exemplary and is not at all intended to restrict the present disclosure, applications thereof, or uses thereof.
- (12) FIG. 1 is a left side view of an electric vehicle (electric automobile) 1 including a vehicle-body structure A according to the present disclosure. As illustrated in FIG. 2, the electric vehicle 1 includes a lower structure 2 and an upper structure 3. In FIG. 1, a front bumper, a rear bumper, front and rear wheels, and so forth are omitted and are illustrated by imaginary lines, and each portion is schematically illustrated. In FIG. 2, in addition to the components omitted in FIG. 1, doors, a bonnet hood, a front fender, window glass, front and rear lighting devices, interior materials, and so forth are omitted, and each portion is schematically illustrated.
- (13) Note that in the description of the embodiment, a vehicle front side will simply be referred to as “front”, a vehicle rear side will simply be referred to as “rear”, a vehicle right side will simply be referred to as “right”, and a vehicle left side will simply be referred to as “left”. A right-left direction of the vehicle is a vehicle width direction.
- (14) As illustrated in FIG. 1, the electric vehicle 1 is a passenger automobile. The electric vehicle 1 may be of any of a sedan type, a hatch-back type, a minivan type, and so forth, and its shape is not particularly limited. As illustrated in FIG. 2, in the electric vehicle 1, a vehicle cabin R1 is formed which serves as a staying space (vehicle cabin inside space) for an occupant. As illustrated in FIG. 1, a front seat (seat) S1 is provided on a front side in the vehicle cabin R1, and a rear seat S2 is provided in rear of the front seat S1 in the vehicle cabin R1. In the rear of the rear seat S2, a trunk R2 is provided in accordance with necessity. The vehicle cabin R1 and the trunk R2 are provided to the upper structure 3. Note that in the vehicle cabin R1, only the front seat S1 may be provided, or a third row seat may be provided in the rear of the rear seat S2.
- (15) Meanwhile, a space (front-side space) in front of the vehicle cabin R1 as a front portion of the electric vehicle 1 can be set as a power chamber R3, for example. That is, as illustrated in FIG. 3, the vehicle-body structure A is provided to the electric vehicle 1 which includes a front-side

traveling motor **M1** installed in a vehicle front portion, a rear-side traveling motor **M2** installed in a vehicle rear portion, batteries **B** supplying electric power to the traveling motors **M1** and **M2**, and a battery casing **10** housing the batteries **B**. The battery casing **10** is disposed below a floor panel **70** described later.

(16) The front-side traveling motor **M1** produces a driving force for driving left and right front wheels **FT**, and a front-side power train **PT1** is configured with only the front-side traveling motor **M1** or with the front-side traveling motor **M1**, a speed reducer, a transmission, and so forth. Further, the rear-side traveling motor **M2** illustrated in FIG. 2 and FIG. 3 produces a driving force for driving left and right rear wheels **RT** (illustrated in FIG. 1), and a rear-side power train **PT2** is configured with only the rear-side traveling motor **M2** or with the rear-side traveling motor **M2**, a speed reducer, a transmission, and so forth.

(17) In the present embodiment, the rear-side traveling motor **M2** is configured to produce a highest output (maximum torque) which is high compared to the front-side traveling motor **M1**, and the rear-side traveling motor **M2** has a larger size than the front-side traveling motor **M1**. Accompanying that, the rear-side power train **PT2** becomes larger than the front-side power train **PT1**. Note that the rear-side traveling motor **M2** may produce a highest output which is low compared to the front-side traveling motor **M1**, or the rear-side traveling motor **M2** and the front-side traveling motor **M1** may produce equivalent highest outputs. Further, only the front-side power train **PT1** may be provided, or only the rear-side power train **PT2** may be provided. Further, for example, in a case of a large-sized vehicle, the front-side traveling motor **M1** and the rear-side traveling motor **M2** are installed which are large compared to a small-sized vehicle.

(18) As illustrated in FIG. 2, the lower structure **2** includes the battery casing **10**, a pair of left and right front side frames **11** and **12** which extend forward in front of the battery casing **10**, and a pair of left and right rear frames **13** and **14** which extend rearward in the rear of the battery casing **10**. The left and right rear frames **13** and **14** extend in a front-rear direction in the vehicle rear portion. A reference numeral **11** denotes the left front side frame, and a reference numeral **12** denotes the right front side frame. Further, a reference numeral **13** denotes the left rear frame, and a reference numeral **14** denotes the right rear frame. In FIG. 2, a lid body **35** (described later) of the battery casing **10** is detached.

(19) In a case of a common electric automobile, a battery casing is often formed as a separate body from a vehicle body and is often detachable from a portion below a floor; however, in the present embodiment, not only the battery casing **10**, by integrating the left and right front side frames **11** and **12** and the left and right rear frames **13** and **14** with the battery casing **10**, but also the front side frames **11** and **12** and the rear frames **13** and **14**, together with the battery casing **10**, are detachable from the upper structure **3**.

(20) Specifically, the electric vehicle **1** of the present embodiment is configured to be capable of being divided, in an up-down direction, into the lower structure **2** having the battery casing **10** and the upper structure **3** forming the vehicle cabin **R1** and the trunk **R2**. Being capable of being divided in the up-down direction means that without using welding, adhesion, or the like, the lower structure **2** is integrated with the upper structure **3** by using fastening members such as bolts, nuts, and screws. Accordingly, because the lower structure **2** can be separated from the upper structure **3** in accordance with necessity when maintenance or repairs are performed after the electric vehicle **1** is passed into the hands of a user, high maintainability is achieved. Note that fastening members used in the following description include bolts, nuts, screws, and so forth.

(21) Here, as a vehicle-body structure of an automobile, a vehicle-body structure of a ladder frame type has been known. In a case of the vehicle-body structure of the ladder frame type, the vehicle-body structure is being capable of being divided, in the up-down direction, into a ladder frame and a cabin, but the ladder frame continuously extends in the front-rear direction and thus mainly receives a collision load in a front collision and a rear collision. In a side collision, the ladder frame only subsidiarily receives a collision load, and the collision load is mainly received by the cabin.

As described above, in the vehicle-body structure of the ladder frame type, usually, different members receive collision loads between the front collision and rear collision and the side collision.

(22) On the other hand, in a case of the electric vehicle **1** of the present embodiment, the lower structure **2** having the front side frames **11** and **12** and the rear frame **13** and **14** and the upper structure **3** are capable of being divided; however, a technical idea of the present embodiment is largely different from the vehicle-body structure of the ladder frame type in related art in the point that in both cases of the front collision and rear collision and the side collision, a collision load is received by the lower structure **2** and the upper structure **3**, and the collision load is capable of being dispersedly absorbed by both of the structures **2** and **3**. In the following, structures of the lower structure **2** and the upper structure **3** will be described in detail.

(23) (Lower Structure)

(24) First, the lower structure **2** will be described. The lower structure **2** includes the front and rear power trains **PT1** and **PT2**, the front wheels **FT**, the rear wheels **RT**, a front suspension apparatus **20**, a rear suspension apparatus **21**, and so forth in addition to the battery casing **10**, the front side frames **11** and **12**, and the rear frames **13** and **14**. Forms of the front suspension apparatus **20** and the rear suspension apparatus **21** are not particularly specified.

(25) A battery unit **BY** is configured with the battery casing **10** and the batteries **B** housed in an internal portion of the battery casing **10**, but in addition to those, for example, a battery cooling device and so forth may be included in the battery unit **BY**.

(26) The battery casing **10** is a large casing which is formed, below the floor panel **70** of the upper structure **3**, to be spanned from a left end portion vicinity to a right end portion vicinity of the floor panel **70** and to be spanned from a front end portion vicinity to a rear end portion vicinity of the floor panel **70**. As described above, the battery casing **10** is disposed in a wide range of a lower region of the floor panel **70**, and it thereby becomes possible to install the battery **B** with a large capacity in the electric vehicle **1**. The battery **B** may be a lithium-ion battery, a solid-state battery, or the like or may be another secondary cell. Further, the battery **B** may be a so-called battery cell or may be a battery pack housing plural battery cells. In the present embodiment, the battery **B** is configured with a battery pack, and plural battery packs are installed in a state where those are aligned in the front-rear direction and the left-right direction.

(27) The battery casing **10** includes a left-side battery frame **30**, a right-side battery frame **31**, a front-side battery frame **32**, a rear-side battery frame **33**, a bottom plate **34**, and the lid body **35** (illustrated in FIG. 5) which covers the batteries **B** from above. Note that FIG. 2 and FIG. 3 illustrate a state where the lid body **35** is detached.

(28) The left-side battery frame **30**, the right-side battery frame **31**, the front-side battery frame **32**, and the rear-side battery frame **33** are configured with an extruded material or the like of an aluminum alloy, for example, but may be configured with an aluminum alloy plate material or a press-formed material of a steel plate as well. The bottom plate **34** can also be configured with an extruded material. In the following description, “extruded material” denotes an extruded material of an aluminum alloy, and “press-formed material” denotes an aluminum alloy plate material or a press-formed material of a steel plate. Further, each member may be configured with a casting, for example.

(29) All of cross-sectional shapes of the left-side battery frame **30**, the right-side battery frame **31**, the front-side battery frame **32**, and the rear-side battery frame **33** in respective orthogonal directions to their longitudinal directions are rectangular shapes. Further, the left-side battery frame **30**, the right-side battery frame **31**, the front-side battery frame **32**, and the rear-side battery frame **33** are all arranged at the same height and extend in generally horizontal directions.

(30) The left-side battery frame **30** and the right-side battery frame **31** are outer-side battery frames which extend in the front-rear direction on vehicle-width-direction outer sides of the batteries **B**. The left-side battery frame **30** is provided to a left-side portion of the battery casing **10** and extends

in the front-rear direction along a left side sill **74**. The left-side battery frame **30** is mounted on the left side sill **74** by fastening members or the like. The right-side battery frame **31** is provided to a right-side portion of the battery casing **10** and extends in the front-rear direction along a right side sill **75**. The right-side battery frame **31** is mounted on the right side sill **75** by fastening members or the like. The left-side battery frame **30** and the right-side battery frame **31** are battery protection frames which protect the batteries **B**.

(31) Further, the front-side battery frame **32** is provided to a front portion of the battery casing **10** and extends in the left-right direction. Further, the rear-side battery frame **33** extends in the left-right direction in a rear portion of the battery casing **10**, that is, in the rear of the batteries **B**. The front-side battery frame **32** and the rear-side battery frame **33** are battery protection frames which protect the batteries **B**.

(32) A left end portion of the front-side battery frame **32** is connected with a front end portion of the left-side battery frame **30**, and a right end portion of the front-side battery frame **32** is connected with a front end portion of the right-side battery frame **31**. A left end portion of the rear-side battery frame **33** is connected with a rear end portion of the left-side battery frame **30**, and a right end portion of the rear-side battery frame **33** is connected with a rear end portion of the right-side battery frame **31**. Consequently, the left-side battery frame **30**, the right-side battery frame **31**, the front-side battery frame **32**, and the rear-side battery frame **33** are members which configure a rack frame formed to surround all of the batteries **B** in a plan view.

(33) Both of the left and right end portions of the front-side battery frame **32** and both of the left and right end portions of the rear-side battery frame **33** are respectively mounted on the left and right side sills **74** and **75** by fastening members or the like. Further, both of the left and right end portions of the rear-side battery frame **33** are connected with the left-side battery frame **30** and the right-side battery frame **31** and are thus mounted on the left and right side sills **74** and **75** via the left-side battery frame **30** and the right-side battery frame **31**. Further, both of the left and right end portions of the rear-side battery frame **33** may directly be mounted on the left and right side sills **74** and **75**.

(34) The bottom plate **34** illustrated in FIG. **4** and so forth extends generally horizontally and is fixed to lower surfaces of the left-side battery frame **30**, the right-side battery frame **31**, the front-side battery frame **32**, and the rear-side battery frame **33**. Further, the lid body **35** illustrated in FIG. **5**, FIG. **6**, and so forth is fixed to upper surfaces of the left-side battery frame **30**, the right-side battery frame **31**, the front-side battery frame **32**, and the rear-side battery frame **33**. In other words, the lid body **35** is mounted on the battery frames **30** to **33**. The rear-side battery frame **33** is protruded to the rear of a rear portion of the lid body **35**.

(35) When the lid body **35** is mounted on the battery frames **30** to **33**, for example, fastening members may be used, or adhesion, welding, or the like may be used. Consequently, a battery housing space **S** housing the batteries **B** (illustrated in FIG. **2**) is marked off and formed with the left-side battery frame **30**, the right-side battery frame **31**, the front-side battery frame **32**, the rear-side battery frame **33**, the bottom plate **34**, and the lid body **35**.

(36) The size of the battery housing space **S** can be changed in accordance with the capacity of the installed batteries **B**. The size of the battery housing space **S** is capable of being easily changed by changing lengths of the left-side battery frame **30**, the right-side battery frame **31**, the front-side battery frame **32**, and the rear-side battery frame **33** and a shape of the bottom plate **34**. For example, in a case where the electric vehicle **1** is a small vehicle which has a short wheelbase and narrow treads, the left-side battery frame **30**, the right-side battery frame **31**, the front-side battery frame **32**, and the rear-side battery frame **33** are made short, the shapes of the bottom plate **34** and the lid body **35** are made small in response to the shortening, and the battery housing space **S** thereby becomes small in accordance with the small vehicle. On the other hand, in a case of a large vehicle, the left-side battery frame **30**, the right-side battery frame **31**, the front-side battery frame **32**, and the rear-side battery frame **33** are made long, the shapes of the bottom plate **34** and the lid

body **35** are made large in response to the elongation, and the battery housing space **S** thereby becomes large in accordance with the large vehicle. In a case where the left-side battery frame **30**, the right-side battery frame **31**, the front-side battery frame **32**, and the rear-side battery frame **33** are configured with the extruded material, the lengths can easily be changed. Further, the bottom plate **34** can also be configured with the extruded material, and its shape can thereby easily be changed.

(37) An upper portion of the battery housing space **S** may be closed by the above lid body **35** or may be closed by the floor panel **70** of the upper structure **3**. In the battery housing space **S**, other than the batteries **B**, a cooling device cooling the batteries **B**, a heating device heating the batteries **B**, and so forth (temperature adjustment devices) can also be provided. Further, electric power of the batteries **B** is supplied to the traveling motors **M1** and **M2** via a control device. In addition, it is possible to charge the batteries **B** via a charging socket, a contactless charger, or the like.

(38) As illustrated in FIG. 2, in the internal portion of the battery casing **10** configuring the battery unit **BY**, as strength members extending in the left-right direction, first to third inside-casing members (inside-unit members) **25A**, **25B**, and **25C** are provided. All of heights of the first to third inside-casing members **25A**, **25B**, and **25C** are the same and are generally the same as the heights of the left-side battery frame **30** and so forth. The inside-casing members **25A**, **25B**, and **25C** may be configured with the extruded material or may be configured with the press-formed material. In the present embodiment, three inside-casing members **25A**, **25B**, and **25C** are provided, but the number of inside-casing members **25A**, **25B**, and **25C** may be increased or decreased in accordance with the dimension of the battery casing **10** in the front-rear direction. The first to third inside-casing members **25A**, **25B**, and **25C** are second members.

(39) The first to third inside-casing members **25A**, **25B**, and **25C** are arranged at distances from each other in the front-rear direction, the first inside-casing member **25A** is in a foremost position, and the third inside-casing member **25C** is in a rearmost position. A lower portion of each of the inside-casing members **25A**, **25B**, and **25C** is fixed to an upper surface of the bottom plate **34**. Further, a left end portion of each of the inside-casing members **25A**, **25B**, and **25C** is fixed to an inner surface (right-side surface) of the left-side battery frame **30**, and a right end portion of each of the inside-casing members **25A**, **25B**, and **25C** is fixed to an inner surface (left-side surface) of the right-side battery frame **31**. In other words, the inside-casing members **25A**, **25B**, and **25C** are members which connect the left-side battery frame **30** and the right-side battery frame **31** together.

(40) In the internal portion of the battery casing **10**, as strength members extending in the front-rear direction, a front central member (inside-unit member) **26** and first to third rear central members (inside-unit members) **27** to **29** are provided. The front central member **26** and the first to third rear central members **27** to **29** are arranged at generally the same heights and are provided at a center of the battery casing **10** in the left-right direction. Lower end portions of the front central member **26** and the first to third rear central members **27** to **29** are mounted on the upper surface of the bottom plate **34**. The front central member **26** and the first to third rear central members **27** to **29** are first members. The front central member **26** and first to third rear central members **27** to **29** and the first to third inside-casing members **25A**, **25B**, and **25C** intersect with each other.

(41) The front central member **26** is arranged between the front-side battery frame **32** and the first inside-casing member **25A**, a front end portion of the front central member **26** is fixed to a central portion of the front-side battery frame **32** in the left-right direction, and a rear end portion of the front central member **26** is fixed to a central portion of the first inside-casing member **25A** in the left-right direction. Consequently, the front-side battery frame **32** is a member which extends so as to connect the front end portions of the left-side battery frame **30** and the right-side battery frame **31** with the front end portion of the front central member **26**.

(42) The first rear central member **27** is arranged between the first inside-casing member **25A** and the second inside-casing member **25B**, a front end portion of the first rear central member **27** is fixed to the central portion of the first inside-casing member **25A** in the left-right direction, and a

rear end portion of the first rear central member **27** is fixed to a central portion of the second inside-casing member **25B** in the left-right direction. Further, the second rear central member **28** is arranged between the second inside-casing member **25B** and the third inside-casing member **25C**, a front end portion of the second rear central member **28** is fixed to the central portion of the second inside-casing member **25B** in the left-right direction, and a rear end portion of the second rear central member **28** is fixed to a central portion of the third inside-casing member **25C** in the left-right direction. Further, the third rear central member **29** is arranged between the third inside-casing member **25C** and the rear-side battery frame **33**, a front end portion of the third rear central member **29** is fixed to the central portion of the third inside-casing member **25C** in the left-right direction, and a rear end portion of the third rear central member **29** is fixed to a central portion of the rear-side battery frame **33** in the left-right direction. Consequently, because the first to third inside-casing members **25A**, **25B**, and **25C** and the front central member **26** and first to third rear central members **27** to **29** are disposed in a lattice manner in the internal portion of the battery casing **10** and are coupled with each other, a reinforcement effect for the battery casing **10** is further enhanced.

(43) When an imaginary straight line extending in the front-rear direction is presumed in a plan view, the positions of the front central member **26** and the first to third rear central members **27** to **29** in the left-right direction are set such that the positions are arranged on the imaginary straight line. In other words, the first to third rear central members **27** to **29** are provided to be positioned on a rearward imaginary extension line of the front central member **26**. Note that the front central member **26** and the first to third rear central members **27** to **29** may be configured with one member which is continuous in the front-rear direction.

(44) As also illustrated in FIG. 5 and FIG. 6, the lower structure **2** includes the rear-side battery frame **33**, the left and right rear frames **13** and **14**, and a vehicle-body-side frame **40**, and those members configure a part of the vehicle-body structure A. The rear frames **13** and **14** can be configured with the extruded material, the press-formed material, or the like, for example. In the present embodiment, because the rear frames **13** and **14** are configured with the extruded material, their cross-sectional shapes in a direction orthogonal to the front-rear direction are generally equivalent from front end portions to rear end portions.

(45) The left and right rear frames **13** and **14** are mounted on the rear-side battery frame **33** configuring the rear portion of the battery casing **10** via the vehicle-body-side frame **40**. In other words, the vehicle-body-side frame **40** extends toward the rear-side battery frame **33** and also extends in the left-right direction along the rear-side battery frame **33**, front portions of the left and right rear frames **13** and **14** are fixed to the vehicle-body-side frame **40**, and the vehicle-body-side frame **40** is coupled with the rear-side battery frame **33**.

(46) Specifically, the vehicle-body-side frame **40** is formed with a member which is integrally shaped by using metal and can also be referred to as cross member because the vehicle-body-side frame **40** extends in the left-right direction. The front portions of the left and right rear frames **13** and **14** are fixed to the vehicle-body-side frame **40** by using welding, adhesion, fastening members, or the like. Metal which configures the vehicle-body-side frame **40** is not particularly limited. For example, aluminum and so forth can be raised, and in this case, the vehicle-body-side frame **40** can be formed by aluminum die-casting.

(47) The left and right rear frames **13** and **14** are mounted on the rear-side battery frame **33** via the vehicle-body-side frame **40**, but the front portions of the rear frames **13** and **14** can be caused to adjoin a rear surface of the rear-side battery frame **33**. Consequently, the rear frames **13** and **14** extend rearward from the rear-side battery frame **33**. Note that the front portions of the rear frames **13** and **14** may slightly be spaced apart rearward from the rear surface of the rear-side battery frame **33**. In this case also, viewing those as the whole, it can be considered that the rear frames **13** and **14** extend rearward from the rear-side battery frame **33**.

(48) The front portion of the left rear frame **13** is arranged to correspond to a section on a left side

of a center of the rear-side battery frame **33** in the left-right direction. Further, the front portion of the right rear frame **14** is arranged to correspond to a section on a right side of the center of the rear-side battery frame **33** in the left-right direction. Accordingly, a distance between the left and right rear frames **13** and **14** becomes a predetermined distance. The distance between the rear frames **13** and **14** is set narrower than a distance between the left-side battery frame **30** and the right-side battery frame **31** of the battery casing **10**. Further, heights of the left and right rear frames **13** and **14** are generally the same.

(49) As illustrated in FIG. **8** to FIG. **10**, the rear-side battery frame **33** has a hollow cross-section in which the cross-section in a direction orthogonal to the longitudinal direction is hollow, and a space is formed in an internal portion. That is, the rear-side battery frame **33** has an upper wall portion **33a** which extends in the left-right direction, a front wall portion **33b** which extends downward from a front edge portion of the upper wall portion **33a** and extends in the left-right direction, a rear wall portion **33c** which extends downward from a rear edge portion of the upper wall portion **33a** and extends in the left-right direction, and a lower wall portion **33d** which extends from a lower edge portion of the front wall portion **33b** to a lower edge portion of the rear wall portion **33c** and extends in the left-right direction. Consequently, the cross-section of the rear-side battery frame **33** has a rectangular shape, but a cross-sectional shape of the rear-side battery frame **33** is not limited to a rectangular shape but may be a complicated polygonal shape. Dimensions of the upper wall portion **33a** and the lower wall portion **33d** in the front-rear direction are set the same, but the dimensions are not limited to this, and one of those may be shorter than the other. Further, dimensions of the front wall portion **33b** and the rear wall portion **33c** in the up-down direction are set the same, but the dimensions are not limited to this, and one of those may be shorter than the other.

(50) In the internal portion of the rear-side battery frame **33**, a first partition wall portion **33e** which marks off the space in the internal portion of the rear-side battery frame **33** in the up-down direction is provided between the upper wall portion **33a** and the lower wall portion **33d**. The first partition wall portion **33e** extends in the front-rear direction from an intermediate portion of the front wall portion **33b** in the up-down direction to an intermediate portion of the rear wall portion **33c** in the up-down direction and also extends in the left-right direction. The first partition wall portion **33e** is generally parallel with the upper wall portion **33a**, and the first partition wall portion **33e** is generally parallel with the lower wall portion **33d**. In the present embodiment, the first partition wall portion **33e** is horizontal but may be inclined. Further, a third partition wall portion, a fourth partition wall portion, or the like may be provided.

(51) In the internal portion of the rear-side battery frame **33**, a second partition wall portion **33f** which marks off the space in the internal portion of the rear-side battery frame **33** in the front-rear direction is provided between the front wall portion **33b** and the rear wall portion **33c**. The second partition wall portion **33f** and the first partition wall portion **33e** are generally orthogonal and are integrated together in a state where an intermediate portion of the second partition wall portion **33f** in the up-down direction intersects with an intermediate portion of the first partition wall portion **33e** in the front-rear direction. The second partition wall portion **33f** extends in the front-rear direction from an intermediate portion of the upper wall portion **33a** in the front-rear direction to an intermediate portion of the lower wall portion **33d** in the up-down direction and also extends in the left-right direction. The second partition wall portion **33f** is generally parallel with the front wall portion **33b**, and the second partition wall portion **33f** is generally parallel with the rear wall portion **33c**. In the present embodiment, the second partition wall portion **33f** is perpendicular but may be inclined.

(52) As illustrated in FIG. **9**, the vehicle-body-side frame **40** includes a vertical plate portion **40a** which extends in the vehicle width direction and the up-down direction along the rear wall portion **33c** of the rear-side battery frame **33** and a lower plate portion **40b** which extends forward from a lower edge portion of the vertical plate portion **40a** along the lower wall portion **33d** of the rear-

side battery frame **33** and which extends also in the vehicle width direction. The vertical plate portion **40a** and the lower plate portion **40b** are respectively fixed to the rear wall portion **33c** and the lower wall portion **33d** of the rear-side battery frame **33** by plural fixation members which will be described later. In such a manner, the vertical plate portion **40a** and the lower plate portion **40b** of the vehicle-body-side frame **40** are respectively fixed to the rear wall portion **33c** and the lower wall portion **33d** of the rear-side battery frame **33**, and mounting rigidity of the vehicle-body-side frame **40** with respect to the rear-side battery frame **33** can thereby be enhanced.

(53) As illustrated in FIG. 4 and FIG. 5, the vehicle-body-side frame **40** includes a left outside rib **40c** which is positioned in a left side area of the left rear frame **13**, a left inside rib **40d** which is positioned in a right side area of the left rear frame **13**, a right outside rib **40e** which is positioned in a right side area of the right rear frame **14**, and a right inside rib **40f** which is positioned in a left side area of the right rear frame **14**. The left outside rib **40c** and the left inside rib **40d** inhibit collapse of the left rear frame **13** in the left-right direction, and the right outside rib **40e** and the right inside rib **40f** inhibit collapse of the right rear frame **14** in the left-right direction. In addition, because the ribs **40c**, **40d**, **40e**, and **40f** are arranged to jut out in the left-right direction of the left and right rear frames **13** and **14**, for example, in a case where an impact load from the rear is exerted on the rear frames **13** and **14**, the impact load can be dispersed to a wide range in the left-right direction by using the ribs **40c**, **40d**, **40e**, and **40f**.

(54) The vehicle-body-side frame **40** is provided with a connecting portion **41** which couples rear end portions of the left outside rib **40c** and the left inside rib **40d** with rear end portions of the right outside rib **40e** and the right inside rib **40f**. The connecting portion **41** extends in the left-right direction, and the connecting portion **41** couples intermediate portions of the left and right rear frames **13** and **14** in the front-rear direction with each other. In other words, in a plan view, a rectangular closed cross-section is configured with the left and right rear frames **13** and **14**, the rear-side battery frame **33**, and the connecting portion **41**.

(55) On the left side and right side as the vehicle-width-direction outer sides of the vehicle-body-side frame **40**, a pair of left and right arm connecting portions (suspension connecting portions) **42** and **43** are respectively provided, with which front portions of left and right suspension arms **21A** configuring a part of the rear suspension apparatus **21** are coupled to be rotatable in the up-down direction. The arm connecting portions **42** and **43** extend to a portion above the connecting portion **41**, and upper portions of the arm connecting portions **42** and **43** are each mounted on the upper structure **3** to be detachable.

(56) In rear portions of the left and right rear frames **13** and **14**, a rear suspension cross member **44** is provided which extends in the left-right direction. The rear portions of the left and right rear frames **13** and **14** are coupled with each other by the rear suspension cross member **44**. On the left side and right side as the vehicle-width-direction outer sides of the rear suspension cross member **44**, a pair of left and right arm connecting portions **44a** and **44b** are respectively provided with which rear portions of the left and right suspension arms **21A** are coupled to be rotatable in the up-down direction. The arm connecting portions **44a** and **44b** extend upward, and upper portions of the arm connecting portions **44a** and **44b** are each mounted on the upper structure **3** to be detachable. Further, in a plan view, a rectangular closed cross-section is configured with the left and right rear frames **13** and **14**, the connecting portion **41**, and the rear suspension cross member **44**.

(57) A subframe **46** is provided in a rear portion of the lower structure **2**. The subframe **46** includes a left member **46a** which is arranged in the rear of the left rear frame **13** and extends in the front-rear direction, a right member **46b** which is arranged in the rear of the right rear frame **14** and extends in the front-rear direction, and a rear member **46c** which couples rear end portions of the left member **46a** and the right member **46b** with each other.

(58) Front portions of the left member **46a** and the right member **46b** of the subframe **46** are fixed to the rear suspension cross member **44**. In other words, the subframe **46** is coupled with the left

and right rear frames **13** and **14** via the rear suspension cross member **44**. As illustrated in FIG. 6, in a connected state, the left and right rear frames **13** and **14** are positioned below the subframe **46**.
(59) (Upper Structure)

(60) Next, the upper structure **3** will be described. As illustrated in FIG. 2, the upper structure **3** includes the floor panel **70**, a dash panel (partition wall portion) **71**, and the pair of left and right side sills **74** and **75**. A reference numeral **74** denotes the left side sill, and a reference numeral **75** denotes the right side sill.

(61) The floor panel **70** configures a floor surface of the vehicle cabin **R1** and is formed with a steel plate or the like which extends in the front-rear direction and extends also in the left-right direction. A space above the floor panel **70** serves as the vehicle cabin **R1**. A roof **80** is provided to an upper portion of the vehicle cabin **R1**. Further, in both of left and right side portions of the upper structure **3**, front openings **3a** and rear openings **3b** are respectively formed. As illustrated in FIG. 1, the front opening **3a** and the rear opening **3b** are capable of being opened and closed by a front door **81** and a rear door **82**, respectively. Note that a front door and a rear door are disposed on a right side of the upper structure **3** to be capable of being opened and closed.

(62) The left and right side sills **74** and **75** are respectively disposed in both of left and right end portions of the floor panel **70** to extend in the front-rear direction. The left end portion of the floor panel **70** is connected with an intermediate portion of the left side sill **74** in the up-down direction, and the right end portion of the floor panel **70** is connected with an intermediate portion of the right side sill **75** in the up-down direction. Upper-side portions of the side sills **74** and **75** are protruded upward from connected sections with the floor panel **70**, and lower-side portions of the side sills **74** and **75** are protruded downward from the connected sections with the floor panel **70**. Because the battery casing **10** is arranged below the floor panel **70**, the battery casing **10** is arranged between the left and right side sills **74** and **75**, and in a vehicle side view, the lower-side portions of the side sills **74** and **75** overlap with the battery casing **10**.

(63) The dash panel **71** is a member which extends in the vehicle width direction and the up-down direction and is for partitioning the vehicle cabin **R1** from the power chamber **R3**. A lower end portion of the dash panel **71** is connected with the front end portion of the floor panel **70**.

(64) FIG. 7 is a bottom view of a rear-side portion of the upper structure **3**, and as illustrated in FIG. 7, the floor panel **70** of the upper structure **3** has a rear panel portion **70A** which configures a floor surface of the trunk **R2**. On both of left and right sides of the rear panel portion **70A**, left and right rear wheel wells **72** and **73** are respectively provided.

(65) The upper structure **3** includes a pair of left and right rear side frames **90** and **91** which extend in the front-rear direction in the vehicle rear portion. The left rear side frame **90** is formed to extend along a left end portion of the rear panel portion **70A** on the vehicle-width-direction inner side of the left rear wheel well **72**. In a rear portion of the left rear side frame **90**, a left crush can **90a** is provided to be protruded to the rear of a rear portion of the rear panel portion **70A**.

(66) The right rear side frame **91** is formed to extend along a right end portion of the rear panel portion **70A** on the vehicle-width-direction inner side of the right rear wheel well **73**. In a rear portion of the right rear side frame **91**, a right crush can **91a** is provided to be protruded to the rear of the rear portion of the rear panel portion **70A**. A bumper reinforcement **92** which extends in the left-right direction is mounted on the rear portion of the left crush can **90a** and the rear portion of the right crush can **91a**. Front-side portions of the left and right rear side frames **90** and **91** are formed to be bent or inclined downward.

(67) Further, as illustrated in FIG. 4 and FIG. 8, the rear-side battery frame **33** of the battery casing **10** extends in the vehicle width direction in front of front portions of the rear side frames **90** and **91**. More specifically, because the battery casing **10** is provided below the floor panel **70**, the rear-side battery frame **33** is arranged below the front portions of the rear side frames **90** and **91**.

(68) The front portion of the left rear side frame **90** is spaced apart, by a prescribed distance, from a rear portion of the left side sill **74** to the rear and to the vehicle-width-direction inner side, and the

front portion of the left rear side frame **90** is not connected with the rear portion of the left side sill **74**. Further, similarly, the front portion of the right rear side frame **91** is spaced apart, by a prescribed distance, from a rear portion of the right side sill **75** to the rear and to the vehicle-width-direction inner side, and the front portion of the right rear side frame **91** is not connected with the rear portion of the right side sill **75**. That is, hypothetically, when a case is presumed where the left rear side frame **90** is connected with the left side sill **74**, the front portion of the rear side frame **90** is elongated forward and is thereby connected with the rear portion of the side sill **74**. In such a case, the front portion of the rear side frame **90** is protruded to the vehicle-width-direction inner side of the rear wheel well **72**. The same applies to the right side, and the front portion of the rear side frame **91** is protruded to the vehicle-width-direction inner side of the rear wheel well **73**. When the left and right rear side frames **90** and **91** are protruded to the vehicle-width-direction inner sides, a dimension, in the left-right direction, of a rear side of the battery casing **10** has to be shortened, and further an installed amount of the batteries **B** is decreased.

(69) In the present embodiment, by not connecting the rear side frames **90** and **91** respectively with the side sills **74** and **75**, a space is formed in which the battery casing **10** is capable of being elongated rearward. Accordingly, because the battery casing **10** can be elongated rearward such that the rear side of the battery casing **10** becomes close to the rear wheel wells **72** and **73** and a long dimension, in the left-right direction, of the rear side of the battery casing **10** can be secured, the installed amount of the batteries **B** can be increased.

(70) For example, as illustrated in FIG. 7, in a case where an imaginary straight line **L1** is presumed which extends forward from a central portion, in the left-right direction, of the front portion of the left rear side frame **90** and an imaginary straight line **L2** is presumed which extends forward from a central portion, in the left-right direction, of the front portion of the right rear side frame **91**, the battery casing **10** (its external shape is indicated by imaginary lines) is provided from a region on the left side of the imaginary straight line **L1** to a region on the right side of the imaginary straight line **L2**. Such a large battery casing **10** is formed, the batteries **B** are thereby installed through an area between vehicle-width-direction outer sides of the imaginary straight lines **L1** and **L2**, and the installed amount of the batteries **B** can further be increased.

(71) Further, in a vehicle side view, the battery casing **10** extends to the rear of the rear portions of the side sills **74** and **75**. Thus, because the rear-side battery frame **33** is positioned in the rear of the rear portions of the side sills **74** and **75**, the batteries **B** are capable of being installed to the rear of the rear portions of the side sills **74** and **75**.

(72) In addition, in a case of not connecting the rear side frames **90** and **91** respectively with the side sills **74** and **75** and where an impact load is exerted from the rear, a problem possibly occurs that there is no direct load transmission path from the rear side frames **90** and **91** to the side sills **74** and **75**; however, in the present embodiment, even when no such load transmission path is present, the impact load from the rear is capable of being absorbed by using the battery casing **10**.

(73) That is, as illustrated in FIG. 8 to FIG. 10, the left and right rear side frames **90** and **91** are coupled with the rear-side battery frame **33** by the vehicle-body-side frame **40**. Specifically, the upper structure **3** includes a rear-side cross member **95** as a vehicle body component which configures a part of the vehicle body, and the rear-side cross member **95** is arranged such that the front portions of the rear side frames **90** and **91** abut the rear-side cross member **95** from the rear and extends in the vehicle width direction. The rear-side cross member **95** includes a front plate portion **95a** which is continuous from the vicinity of a front portion of the left wheel well **72** to the vicinity of a front portion of the right wheel well **73** and extends in the up-down direction and the left-right direction, a rear plate portion **95b** which is arranged to be spaced apart rearward from the front plate portion **95a** and extends in the up-down direction and the left-right direction, and a lower plate portion **95c** which extends from a lower end portion of the front plate portion **95a** to a lower end portion of the rear plate portion **95b**. A dimension of the rear plate portion **95b** in the up-down direction is set longer than a dimension of the front plate portion **95a** in the up-down

direction, and an upper end portion of the rear plate portion **95b** is positioned above an upper end portion of the front plate portion **95a**. Further, inclination angles of the front plate portion **95a** and the rear plate portion **95b** are set such that a distance between the front plate portion **95a** and the rear plate portion **95b** in the front-rear direction becomes wider toward a higher position.

(74) The upper end portion of the front plate portion **95a** and the upper end portion of the rear plate portion **95b** are joined to a lower surface of the floor panel **70**, and a closed cross-section is formed with the front plate portion **95a**, the rear plate portion **95b**, the lower plate portion **95c**, and the floor panel **70**. Accordingly, rigidity of the floor panel **70** can be enhanced.

(75) The front portions of the rear side frames **90** and **91** abut the rear plate portion **95b** of the rear-side cross member **95** from the rear and are joined to the rear plate portion **95b** and the lower plate portion **95c**. Because a dimension of the rear plate portion **95b** in the up-down direction is long compared to the front plate portion **95a**, a wide joining area of the rear side frames **90** and **91** with respect to the rear-side cross member **95** can be secured.

(76) As illustrated in FIG. **8**, the rear-side battery frame **33** is disposed below the rear-side cross member **95**, and the lower plate portion **95c** of the rear-side cross member **95** is formed to extend along an outer surface (upper surface) of the upper wall portion **33a** of the rear-side battery frame **33**. The rear-side cross member **95** and the rear-side battery frame **33** are fastened and fixed together in the up-down direction. Accordingly, the rear-side battery frame **33** is set to a state where that is mounted on the rear-side cross member **95** to be detachable.

(77) Specifically, as illustrated in FIG. **9**, a lower surface of a cylindrical nut **N1** which extends in the up-down direction is fixed to an upper surface of the lower plate portion **95c** of the rear-side cross member **95**. Because the nut **N1** is disposed in the rear-side cross member **95**, an inner space of the rear-side cross member **95** can efficiently be used.

(78) As illustrated in FIG. **10**, plural nuts **N1** are provided at distances from each other in the left-right direction. Plural reinforcement plates **95d** corresponding to the number of nuts **N1** are provided in an internal portion of the rear-side cross member **95**. The reinforcement plate **95d** has a through hole **95e** through which an upper portion of the nut **N1** passes. Because an outer peripheral surface of the nut **N1** is welded to a peripheral edge portion of the through hole **95e** in a state where the upper portion of the nut **N1** passes through the through hole **95e**, fixing strength of the nut **N1** can be enhanced. A front-side portion of the reinforcement plate **95d** is formed along an inner surface of the front plate portion **95a** and is welded to the front plate portion **95a**. A rear-side portion of the reinforcement plate **95d** is formed along an inner surface of the rear plate portion **95b** and is welded to the rear plate portion **95b**. Consequently, a state is established where the front plate portion **95a** and the rear plate portion **95b** are coupled together by the reinforcement plate **95d**.

(79) In the first partition wall portion **33e** of the rear-side battery frame **33**, an insertion hole **33g** is formed through which a bolt **B1** is inserted from below. The same number of insertion holes **33g** as the number of nuts **N1** are provided and formed at the same distances as the distances among the nuts **N1**. The nut **N1** is arranged directly above the insertion hole **33g**. The bolt **B1** inserted through the insertion hole **33g** passes through the upper wall portion **33a** of the rear-side battery frame **33** and is screwed into the nut **N1** from below while passing through the lower plate portion **95c**. Insertion holes through which the bolts **B1** are inserted are formed in the upper wall portion **33a** of the rear-side battery frame **33**, and insertion holes through which the bolts **B1** are inserted are also formed in the lower plate portion **95c** of the rear-side cross member **95**.

(80) Hole portions **33h** are formed in the lower wall portion **33d** of the rear-side battery frame **33**. The hole portion **33h** is for allowing the bolt **B1** to pass through and is formed to open in an outer surface (lower surface) of the lower wall portion **33d** of the rear-side battery frame **33** and to be connected with a space in an internal portion of the rear-side battery frame **33**. A shape of the hole portion **33h** is a circular shape but is not limited to this. Further, a diameter of the hole portion **33h** is set larger than a maximum outer diameter of a head portion of the bolt **B1**, and the bolt **B1** is

enabled to be easily inserted.

(81) The same number of hole portions **33h** as the number of nuts **N1** are also provided and formed at the same distances as the distances among the nuts **N1**. In a state where shaft portions of the bolts **B1** are directed upward, the bolts **B1** are inserted into the space in the internal portion of the rear-side battery frame **33** through the hole portions **33h** and are screwed into the respective nuts **N1**. Accordingly, the rear-side cross member **95** and the rear-side battery frame **33** can be fastened and fixed together at plural parts.

(82) The head portion of the bolt **B1** enters the internal portion of the rear-side battery frame **33** and abuts, from below, a peripheral edge portion of the insertion hole **33g** in the first partition wall portion **33e**.

(83) Accordingly, because the head portion of the bolt **B1** exerts a fixing force, upward from below, on the peripheral edge portion of the insertion hole **33g** in the first partition wall portion **33e**, the rear-side battery frame **33** can be fixed to the rear-side cross member **95** in the internal portion of the rear-side battery frame **33**. In other words, the bolt **B1** is a fixation member which is inserted into the space of the internal portion of the rear-side battery frame **33** through the hole portion **33h** and is capable of fixing the rear-side battery frame **33** to the rear-side cross member **95** as another component than the vehicle-body-side frame **40** in the internal portion of the rear-side battery frame **33**. Note that the fixation member is not limited to a bolt but may be a fastening member such as a screw or a rivet, for example.

(84) The lower plate portion **40b** of the vehicle-body-side frame **40** is formed to extend along the outer surface of the rear-side battery frame **33** in which the hole portions **33h** are formed (the outer surface of the lower wall portion **33d**). Consequently, the hole portions **33h** can be covered from below by the lower plate portion **40b**. In the present embodiment, the lower plate portion **40b** serves as a cover portion which covers the hole portions **33h** from the outside of the rear-side battery frame **33**. The lower plate portion **40b** extends to an area in front of front portions of the hole portions **33h** and can thus completely cover the hole portions **33h**.

(85) (Fastening Structure of Vehicle-Body-Side Frame)

(86) Next, a description will be made about a fastening structure of the vehicle-body-side frame **40** to the rear-side battery frame **33**. As illustrated in FIG. 9, to an inner surface (upper surface) of the lower wall portion **33d** of the rear-side battery frame **33**, plural nuts **N2** are fixed at distances from each other in the vehicle width direction. An axis line of the nut **N2** extends in the up-down direction and is parallel with an axis line of the nut **N1**.

(87) The lower plate portion **40b** of the vehicle-body-side frame **40** is arranged on an outer surface (lower surface) of the lower wall portion **33d** of the rear-side battery frame **33**, and plural bolts **B2** which are screwed into the nuts **N2** pass through, from below, the lower wall portion **33d** of the rear-side battery frame **33** and the lower plate portion **40b** of the vehicle-body-side frame **40**. The bolt **B2**, which passes through, from below, the lower wall portion **33d** of the rear-side battery frame **33** and the lower plate portion **40b** of the vehicle-body-side frame **40**, is screwed into the nut **N2** from below. The bolts **B2** are respectively screwed into the nuts **N2**, and the vehicle-body-side frame **40** and the rear-side battery frame **33** can thereby be fastened together at plural parts.

(88) Further, plural nuts **N3** are fixed to an inner surface (front surface) of the rear wall portion **33c** of the rear-side battery frame **33**. An axis line of the nut **N3** extends in the front-rear direction and is orthogonal to the axis lines of the nut **N1** and nut **N2** in a side view. The plural nuts **N3** are provided at distances from each other in the vehicle width direction and are respectively provided to the upper-side portion and also to the lower-side portion of the rear wall portion **33c** of the rear-side battery frame **33**. The vertical plate portion **40a** of the vehicle-body-side frame **40** is arranged on an outer surface (rear surface) of the rear wall portion **33c** of the rear-side battery frame **33**, and plural bolts **B3** which are screwed into the nuts **N3** pass through the rear wall portion **33c** of the rear-side battery frame **33** and the vertical plate portion **40a** of the vehicle-body-side frame **40**. The bolt **B3**, which passes through, from below, the rear wall portion **33c** of the rear-side battery frame

33 and the vertical plate portion **40a** of the vehicle-body-side frame **40**, is screwed into the nut **N3** from the rear. The bolts **B3** are respectively screwed into the nuts **N3**, and the vehicle-body-side frame **40** and the rear-side battery frame **33** can thereby be fastened together at plural parts.

(89) (Working Effects of Embodiment)

(90) As described above, in the present embodiment, the rear-side battery frame **33** which protects the batteries **B** has a hollow cross-section and thus becomes a member which is highly rigid and light. Further, because the vehicle-body-side frame **40** which extends along the rear-side battery frame **33** is coupled with the rear-side battery frame **33** at plural parts, this also enhances rigidity of the rear-side battery frame **33**.

(91) Furthermore, because the rear-side battery frame **33** is, in its internal portion, fixed to the rear-side cross member **95** by the bolts **B1** separately from the vehicle-body-side frame **40**, compared to a case where the rear-side battery frame **33** is fixed to the rear-side cross member **95** in a lower portion of the rear-side battery frame **33**, the distance between a portion (first partition wall portion **33e**) on which fixing forces of the bolts **B1** are exerted in the rear-side battery frame **33** and the rear-side cross member **95** becomes short. Accordingly, mounting rigidity of the rear-side battery frame **33** on the vehicle body is enhanced.

(92) Further, the hole portions **33h** for inserting the bolts **B1** are formed in the rear-side battery frame **33**, but the hole portions **33h** can be covered by the lower plate portion **40b** of the vehicle-body-side frame **40** from the outside. Accordingly, entrance of foreign objects into the internal portion of the rear-side battery frame **33** can be inhibited. Further, because the hole portions **33h** open in a lower surface of the vehicle body, this might lead to lowering of aerodynamic performance in traveling in a state where the hole portions **33h** are not covered; however, in the present embodiment, because the hole portions **33h** are covered by the lower plate portion **40b**, lowering of aerodynamic performance in traveling can be avoided.

(93) The above-described embodiment is merely an example in all respects and is not to be construed in a limited manner. Furthermore, all modifications and changes belonging to the equivalent scope of the claims are included in the scope of the present disclosure.

(94) [Industrial Applicability]

(95) As described in the foregoing, a vehicle-body structure according to the present disclosure can be provided to an electric vehicle, for example.

Claims

1. A vehicle-body structure for an electric vehicle which includes a traveling motor and in which a battery supplying electric power to the traveling motor is disposed below a floor panel, the vehicle-body structure comprising: a battery protection frame that protects the battery and has a hollow cross-section; a vehicle-body-side frame that extends toward the battery protection frame and is coupled with the battery protection frame; and a fixation member that is inserted into a space of an internal portion of the battery protection frame through a hole portion opening in an outer surface of the battery protection frame and connected with the space, wherein the fixation member fixes the battery protection frame to a vehicle body component as another component than the vehicle-body-side frame in the internal portion of the battery protection frame, wherein the battery protection frame is disposed below the vehicle body component, wherein in the internal portion of the battery protection frame, a partition wall portion that marks off the space in an up-down direction is provided between an upper wall portion and a lower wall portion of the battery protection frame, and wherein the fixation member exerts a fixing force on the partition wall portion.

2. The vehicle-body structure according to claim 1, wherein the vehicle-body-side frame is provided with a cover portion that is formed to extend along the outer surface, in which the hole portion is formed, in the battery protection frame and that covers the hole portion from an outside of the battery protection frame.

3. The vehicle-body structure according to claim 1, wherein the vehicle-body-side frame includes a suspension connecting portion with which at least a part of a suspension apparatus is coupled.
 4. The vehicle-body structure according to claim 3, wherein the battery protection frame extends in a vehicle width direction, and the vehicle-body-side frame includes a lower plate portion that extends in the vehicle width direction along a lower wall portion of the battery protection frame and is fixed to the lower wall portion and a vertical plate portion that extends in the vehicle width direction along one wall portion of the battery protection frame in a vehicle front-rear direction and is fixed to the one wall portion.
 5. The vehicle-body structure according to claim 4, wherein the hole portion is formed in the lower wall portion of the battery protection frame, and the hole portion is covered from below by the lower plate portion of the vehicle-body-side frame.
 6. The vehicle-body structure according to claim 1, wherein the battery protection frame extends in a vehicle width direction, and the vehicle-body-side frame includes a lower plate portion that extends in the vehicle width direction along a lower wall portion of the battery protection frame and is fixed to the lower wall portion and a vertical plate portion that extends in the vehicle width direction along one wall portion of the battery protection frame in a vehicle front-rear direction and is fixed to the one wall portion.
 7. The vehicle-body structure according to claim 2, wherein the vehicle-body-side frame includes a suspension connecting portion with which at least a part of a suspension apparatus is coupled.
 8. The vehicle-body structure according to claim 2, wherein the battery protection frame extends in a vehicle width direction, and the vehicle-body-side frame includes a lower plate portion that extends in the vehicle width direction along a lower wall portion of the battery protection frame and is fixed to the lower wall portion and a vertical plate portion that extends in the vehicle width direction along one wall portion of the battery protection frame in a vehicle front-rear direction and is fixed to the one wall portion.
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